

PROJECT SYNOPSIS

1. Title

Working Project Title: Pregnancy Complication Prediction Using Machine Learning

Team Details

Student Name	College Name	Branch	Mobile No	Mail ID	Role
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Program Track and Domain

Program Track (Engg-HUB / Engg-Spoke / Science-HUB / Science-Spoke / Commerce-Spoke)	Domain (Analytics / Machine Learning / Deep Learning / GenAI)
Science-Spoke	Machine Learning

Guide Details

Guide Name and Designation: Prajapati Jyoti Harinaraynbhai

2. Problem Statement

Pregnancy complications can create significant risks for both the mother and baby, and early identification of possible complications can support timely monitoring and medical attention. This project focuses on developing a machine learning system that can classify whether a pregnancy record indicates a pregnancy complication or no pregnancy complication. The available dataset contains clinical and pregnancy-related information, and the project aims to use these attributes to build and compare classification models. The proposed system is intended to provide a convenient preliminary prediction tool through a Streamlit interface.

3. Objectives

- To understand and analyze clinical and pregnancy-related data.
- To perform data cleaning, exploratory data analysis and preprocessing.
- To develop machine learning classification models for pregnancy complication prediction.
- To implement and compare Logistic Regression, Decision Tree, Random Forest and XGBoost models.
- To evaluate the models using Accuracy, Precision, Recall, F1-Score and ROC-AUC.

- To identify an effective predictive model based on the experimental results.
- To save the trained model and develop a user-friendly Streamlit prediction application.

4. Methodology / Implementation Plan

The project will follow a structured machine learning workflow. The pregnancy dataset will first be loaded and understood, followed by data-quality checks such as missing-value and duplicate analysis. Exploratory Data Analysis will be performed to understand feature distributions, target distribution and relationships among variables. The data will then be prepared for machine learning, including an 80:20 stratified train-test split and scaling where required. Logistic Regression, Decision Tree, Random Forest and XGBoost will be trained and compared. Model performance will be evaluated using Accuracy, Precision, Recall, F1-Score and ROC-AUC, along with confusion matrix and ROC-curve analysis. The selected model will be saved and integrated into a Streamlit-based application where users can enter clinical and pregnancy-related parameters and obtain a predicted risk class.

Overall Workflow:

Dataset Collection → Data Understanding → Data Cleaning → EDA → Preprocessing → Train/Test Split → Model Training → Model Evaluation → Model Saving → Streamlit Deployment

Algorithms / Models:

- Logistic Regression
- Decision Tree
- Random Forest
- XGBoost

Tools / Technologies:

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Scikit-learn
- Matplotlib/Seaborn
- Joblib
- Streamlit

5. Resources and Timeline

A. Hardware & Software Requirements

- Laptop/PC with sufficient RAM and storage for Python and machine learning development.
- Operating System: Windows/Linux/macOS.

- Programming Language: Python.
- Development Tools: Jupyter Notebook and VS Code.
- Libraries/Frameworks: Pandas, NumPy, Scikit-learn, Matplotlib/Seaborn and Streamlit.

B. Dataset

Dataset availability: Available. The project material describes a dataset containing 500 records and 17 columns, including 16 input features and the target variable Pregnancy_Complication. Target classes are 0 = No Pregnancy Complication and 1 = Pregnancy Complication.

Dataset source/link: Not specified in the provided project materials; add the original source link if required.

C. Project Timeline

Phase	Task	Tentative Completion Date
Phase 1	Dataset collection and understanding	Week 1
Phase 2	Data cleaning, EDA and visualization	Week 2
Phase 3	Data preprocessing and train-test split	Week 3
Phase 4	Model development and comparison	Week 4
Phase 5	Performance evaluation and interpretation	Week 5
Phase 6	Model saving and Streamlit deployment	Week 6
Phase 7	Testing, documentation and final demo	Week 7

6. Expected Outcomes / Conclusion

The expected outcome is a functional machine learning system capable of classifying pregnancy records into no pregnancy complication or pregnancy complication classes. The project will provide a comparison of multiple machine learning algorithms and their evaluation using standard classification metrics. The project material reports that the optimized XGBoost model achieved 86% accuracy and 0.9531 ROC-AUC in the observed run. A Streamlit-based prototype will provide a simple interface for entering patient-related parameters and viewing the model prediction. The system is intended as a machine learning-based preliminary prediction tool and not as a replacement for professional medical diagnosis.

7. References

- Project dataset and accompanying project materials provided for the Pregnancy Complication Prediction project.
- Scikit-learn documentation for machine learning classification and model evaluation.
- Python documentation and Jupyter Notebook resources.
- Streamlit documentation for application deployment.